

SETS...

SUB-TOPICS

- Definition of Set
- Types of Set :
 - ⇒ Roster form
 - ⇒ Set Builder form
- Singleton Set (Only One element)
- finite infinite Set
- Union intersection Diff of Set.
- Subset
- Proper Subset improper Subset.
- Equal Set
- Disjoint Set.
- Universal Set
- Complementary Set. denoted by (A^c)
- Power Set $P(A) \Rightarrow$ if A has n element then $P(A) = 2^n$
- Cartesian Product
 - Null Set or Void Set.

$$n(A \cap B') = n(A \cup B) - n(A \cup B)$$

$$n(A \cup B') = n(A \cap B) = n(A \cup B) - n(A \cap B)$$

$$n(A \Delta B) = n[(A - B) \cup (B - A)] = n[(A \cap B') \cup (A' \cap B)]$$

**

$$= n(A) + n(B) - 2n(A \cap B)$$

• Law

$$\rightarrow \text{Identity } A \cup \phi = A \text{ \& } A \cap U = A$$

$$\rightarrow \text{Commutative } A \cup B = B \cup A \text{ \& } A \cap B = B \cap A$$

$$\rightarrow \text{Associative } (A \cup B) \cup C = A \cup (B \cup C)$$

$$\rightarrow \text{Distributive } A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\rightarrow \text{De Morgan's law } (A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

FORMULAS

: Total NO. of Subsets Possible for Particular Element is 2^{n-1}

$$: n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

$$: n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

: Well defined collection of distinct Obj. is called as Set

Useful Prop:

$$1) A \cup U = U$$

$$2) A \cap \phi = \phi$$

$$3) \phi^c = U$$

$$4) U^c = \phi$$

Py & Jee.

$$\boxed{n(A \Delta B) = \phi \text{ then} \\ = n(A) + n(B) - 2n(A \cap B)} \quad \checkmark$$

$$: n(A \cup B) = \phi \Rightarrow n(A \cup B) = n(A) + n(B)$$

$$: n(A \cap B) = \phi \Rightarrow n(A \cup B) = n(A) + n(B)$$

$$* * n(A - B) = n(A) - n(A \cap B)$$

$$= n(A \cup B) - n(B)$$

: If $n(A) = m$

then NO. of Subsets Proper is $2^m - 1$

LESSON-3

SETS!

Sets:

A well-defined collection of Objects.

Representation of Set:

- i. Roster form.
- ii. Set Builder

Roster form:

Here the Set is described by listing each Member of the Set Separated by Commas within Curly brackets.

Ex:

Representation of All Vowels of Eng Alphabets.

$$A = \{a, e, i, o, u\}$$

Set Builder form.

Here a Set is described by Specifying the Characterizing Properties of its elements ...

Ex:

$$A = \{2, 3, 5, 7\}$$

$$A = \{x : x \text{ is a Prime less than } 10\}$$

Types of SetsEmpty Set : (Φ)

- A Set which contain no Elements
- Denoted by ϕ (or) $\{\}$.

Ex: $A = \{x: x \text{ is a natural NO. less than } 1\}$
 $A = \{\}$

Single ton Set:

A Set contain Only One Element

Ex:

$A = \{x: x \text{ is an Even Prime}\}$

Finite Set:

A Set contains finite NO. of Elements

Ex: $A = \{x: x \text{ is a factor of } 24\}$

Infinite Set:

A Set contains infinite NO. of Elements.

Ex:

$A = \{x: x \text{ is a real Number}\}$

Equal Set:

A Sets whose Elements contain Same

$A = \{x: x \text{ is letter of ALLOY}\}$

$B = \{x: x \text{ is letter of LOYAL}\}$

Equivalent Set:

NO. of Elements are Same.

"Cardinality of Set":

Number of Element of A is called Cardinality of A

$n(A) = \text{NO. of Elements in Set A}$

• for finite sets only!

"Subset":

Let A and B be two sets if every element of B is an element of A then B is called subset of A!

A!

Notation: $B \subseteq A$ B is subset of A

$B \not\subseteq A$ B is not a subset of A.

- A subset B of a set A is called proper subset of A is B is contained in A but $A \neq B$
- Here we say A is supersubset of B!
- Total no. of subsets of finite set containing m elements is 2^m .

Example:

$$A = \{1, 2, 3\}$$

Possible subsets: $\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\}$

If $n(A) = m$:

then:

No. of non-empty subsets of A = $2^m - 1$

No. of proper subsets of A = $2^m - 1$

No. of non-void proper subsets of A = $2^m - 2$.

: An element can't be subset...

Power set of A set:

- Denoted by $P(A)$
- Collection of all subsets of A is

Called Power set of A!

3. if A is ϕ $B = P(P(P(A)))$

then which of following are correct?

A) $\{ \} \in B$

$$P(A) = \{ \{ \} \}$$

B) $\{ \{ \} \} \in B$

$$P(P(A)) = \{ \{ \{ \} \}, \{ \{ \{ \} \} \} \}$$

C) $\{ \{ \{ \} \} \} \in B$

$$P(P(P(A)))$$

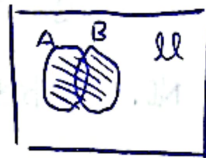
D) $\{ \} \in B$

$$= \{ \{ \{ \{ \{ \} \} \}, \{ \{ \{ \{ \{ \} \} \} \}, \{ \{ \{ \{ \{ \{ \} \} \} \} \} \}$$

Universal Set:

• Super Set of Every Elements

• Denote $\mathcal{U}$



Operations on Sets:

i. Union of Sets:

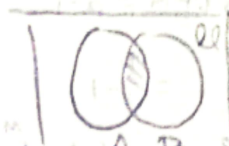
The Union of two sets $A \cup B$

ii. Intersection of Sets:



• $A \cap B$

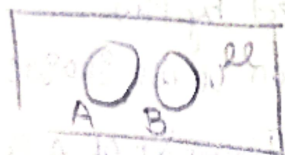
• Set of all Element which are common.



iii. Disjoint Sets:

• $A \cap B = \phi$

• No Common Elements



Min = Max Values of $A \cup B, A \cap B$:

let A, B two sets containing m, n elements respectively then.

(if $n(U) > m+n$)

i. $n(A \cup B)_{\text{Max}} = n(A) + n(B)$

ii. $n(A \cup B)_{\text{Min}} = \max\{m, n\}$

iii. $n(A \cap B)_{\text{Max}} = \min\{m, n\} = (A \cap B) \cup (B \cap A)$

iv. $n(A \cap B)_{\text{Min}} = 0$

$n(A \cap B) = n(A) + n(B) - n(A \cup B)$

Q. Survey of 63% Americans like cheese whereas 76% of people like apples. If $x\%$ of Americans like both cheese and apples then

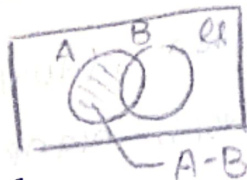
Sol. $n(A \cap B) = x$

$x = 63 + 76 - n(A \cup B)_{\text{Max}}$

$= 139 - 100$

$= 39 \%$

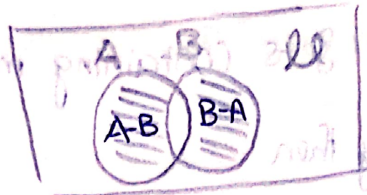
Difference of Sets:



- All elements of A which do not belong to B .

$A - B = A - A \cap B$

Symmetric difference:



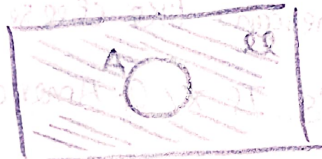
- denoted by $A \Delta B$
- let A, B two sets
then symmetric difference $(A-B) \cup (B-A)$

$$(A-B) \cup (B-A) = (A \cup B) - (A \cap B)$$

Complement of a set:

$$A \cap A^c = \phi$$

$$A \cup A^c = U$$



$$(A')' = A$$

$$U' = \phi$$

$$\phi' = U$$

$$i. n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$ii. n(A \cap B) = \phi \text{ then } n(A \cup B) = n(A) + n(B)$$

$$iii. (A \cap B)' = \phi \text{ then } n(A \cup B) = n(A) + n(B)$$

$$n(A-B) = n(A) - n(A \cap B) = n(A \cup B) - n(B)$$

$$iv. n(A \Delta B) = n(A) + n(B) - 2n(A \cap B)$$

$$v. n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$$